

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2105

Dooville 230 kV
July 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2105. The Interconnection Customer proposes a 321.9 MW solar photovoltaic facility interconnecting at the Dooville 230 kV substation in the Indiana area. The request was received on June 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 694 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,017,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Lonetree 500 kV/230 kV Transformer Bank: 116.8% loading (1,081 MVA) following the outage of the E Methuen-Lonetree 500.0 kV; DFAX = 8.3%.

Murdock-Sabanera 230.0 kV: 100.6% loading (1,527 MVA) following the outage of the Henry County Substation-E Methuen 500.0 kV; DFAX = 7.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lonetree 500 kV/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$29,612,000	6% (\$1,809,000)
Murdock-Sabanera 230.0 kV	Reconductor line	\$10,936,000	2% (\$208,000)
	Total allocated to project		\$2,017,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.